



# Scientific Prospects of the POLAR-2 Mission

Frontier Research in Astrophysics - V  
Mondello, Italy – 19<sup>th</sup> June 2026

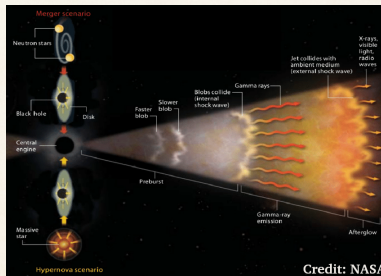
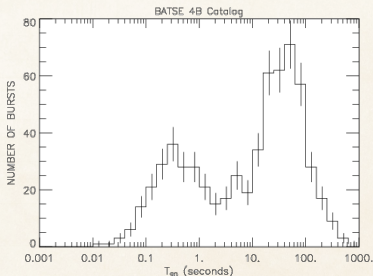
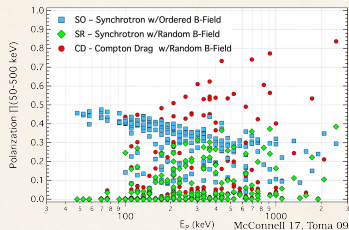
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- Bright and short transient event in  $\gamma$  band followed by an afterglow (in all wavelengths)
- Extragalactic, 2 categories: short (from BNS) and long (from SN)
- Polarization brings a better understanding of the jet and magnetic field structures



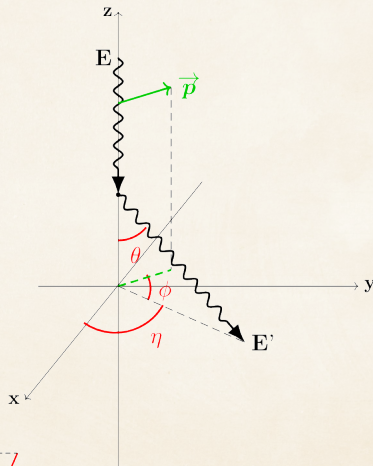
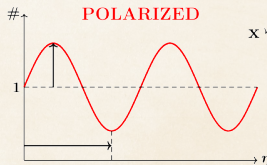
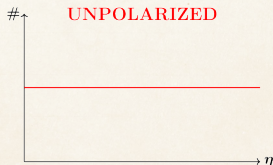
|                                                           |                                                                                              |                  |              |              |
|-----------------------------------------------------------|----------------------------------------------------------------------------------------------|------------------|--------------|--------------|
| Structured Jets<br>$\theta_{\text{obs}} \gtrsim \theta_c$ | 7%-18%                                                                                       | 38%-52%          | 5%-15%       | 10%-25%      |
|                                                           | 0%                                                                                           | 0%-50%           | 0%           | 0%           |
|                                                           | 4%-28%                                                                                       | 10%-60%          | 4%-20%       | 5%-50%       |
|                                                           | $B_{\perp}$ (rand)                                                                           | $B_{\text{tor}}$ | Photospheric | Compton Drag |
|                                                           | Synchrotron                                                                                  |                  |              |              |
|                                                           | Top-Hat Jets<br>$\theta_{\text{obs}} \ll \theta_c$<br>$\theta_{\text{obs}} \approx \theta_j$ |                  |              |              |

Compton scattering can be used to determine the polarization of a source:

- Azimuthal scattering angle distribution provides information on polarization degree and angle
- So-called modulation curved, parameterized by the Klein-Nishina cross-section:

$$\frac{d\sigma}{d\Omega} = \frac{r_e^2}{2} \left( \frac{E'}{E} \right)^2 \left[ \frac{E'}{E} + \frac{E}{E'} - 2 \sin^2(\theta) \cos^2(\phi) \right]$$

- Relative amplitude  $\leftrightarrow$  PD, phase  $\leftrightarrow$  PA



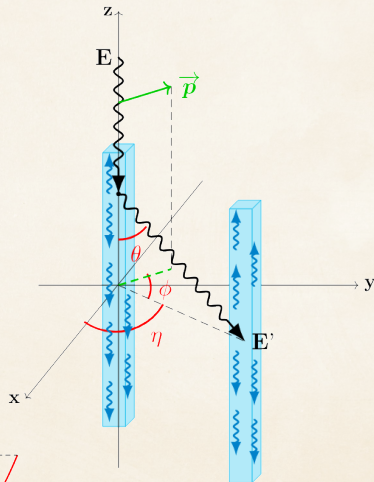
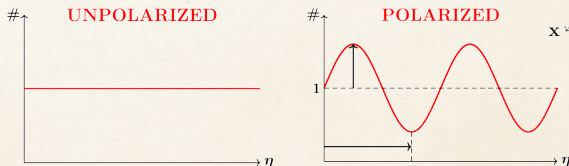


Compton scattering can be used to determine the polarization of a source:

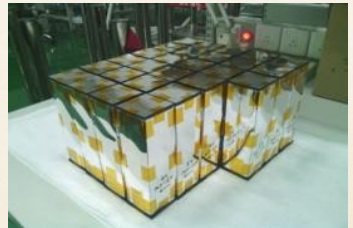
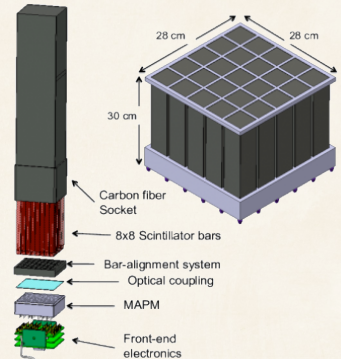
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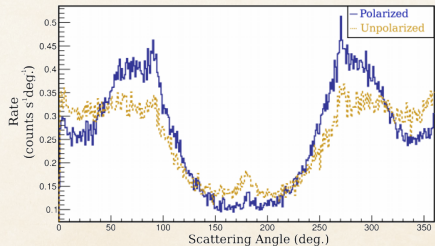
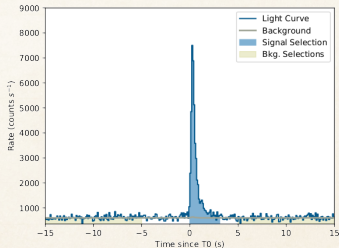
- Relative amplitude  $\leftrightarrow$  PD, phase  $\leftrightarrow$  PA
- **A segmented array of scintillators can be used to measure the scattering angle distribution (aka modulation curve)**



- POLAR was a dedicated gamma polarimeter composed of a  $40 \times 40$  scintillator array
- Divided in  $5 \times 5$  modules each made of 64 plastic scintillator bars ( $176 \times 5.8 \times 5.8 \text{ mm}^3$ , EJ-248M), each module being readout by Multi-Anode PMTs
- Optimized for Compton scattering in the 50-500keV range thanks to its low-Z scintillators
- 30kg instrument, half-sky FoV,  $\sim 300 \text{ cm}^2$  effective area at 400 keV
- Design described in Produit et al. 2018 (DOI: 10.1016/j.nima.2017.09.053)
- Launched in Sept 2016 on the Tiangong-2 Chinese space lab for 6 months of operation

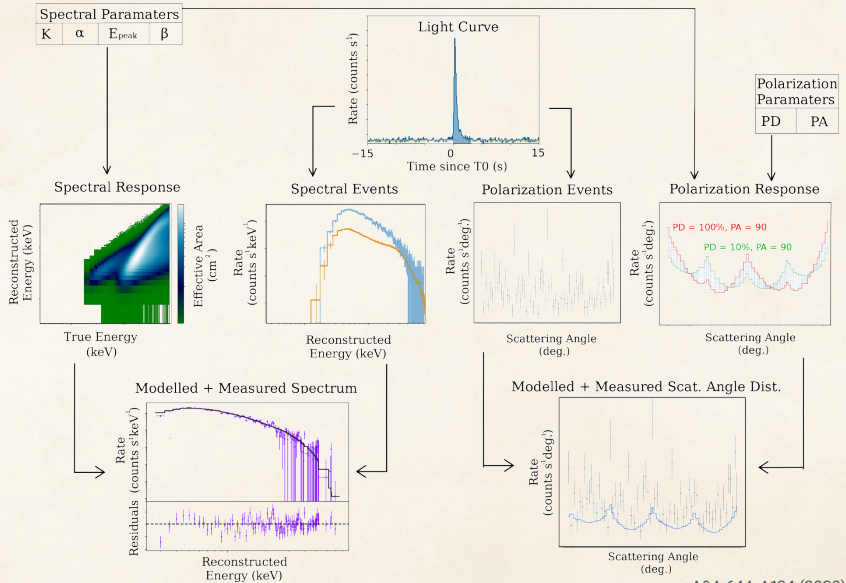


- Typical measured light and modulation curves shown below, complex modulation curve structure due to well-understood instrumental/geometrical effects
- POLAR detected 55 GRBs in 6 months of operation, 14 of which had enough statistics to be analyzed → joint spectral/polarization analysis with Fermi-GBM or Swift-BAT data using 3ML spectral fitting framework ([github.com/threeML](https://github.com/threeML)) and development of a polarization fitting plugin ([github.com/grburgess/polarpy](https://github.com/grburgess/polarpy))



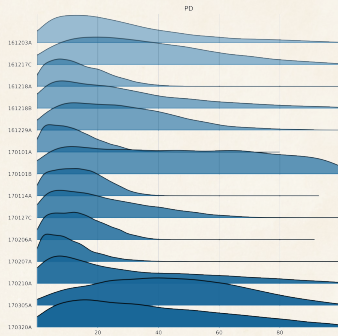
A&A 644, A124 (2020)

# Joint polarization-spectral GRB analysis method

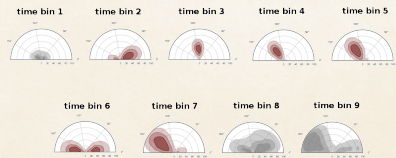


A&A 644, A124 (2020)

- Catalog of 14 GRBs analysed, results show a low or null polarization degree (excluding synchrotron emission models from toroidal magnetic field, compatible with photospheric emission model and other synchrotron models)
- Time resolved analysis show a hint of quickly evolving polarization angle that washes out polarization degree on time integrated analysis  $\Rightarrow$  need more statistics to make proper time resolved analysis
- No significant polarization energy-dependence  $\rightarrow$  **PoS(ICRC2023)619**
- We need more statistics in order to perform temporal and energy resolved analysis, with lowered energy threshold to probe emission models, and with bigger effective area and longer mission operation to get a larger catalog  $\rightarrow$  **the POLAR-2 mission**

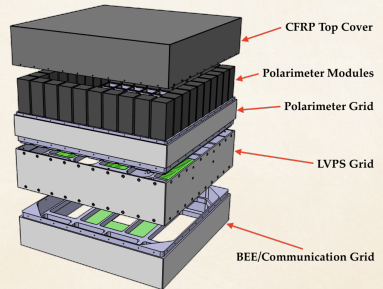
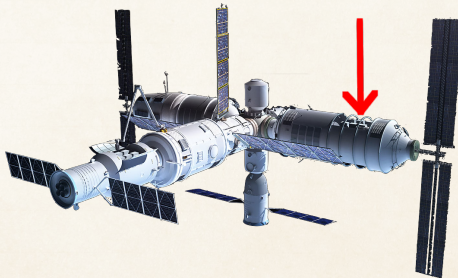


A&A 644, A124 (2020)



A&A 627, A105 (2019)

- Large scale GRB polarimeter based on POLAR legacy
- 4 times bigger than POLAR (from 25 to 100 polarimeter modules), 10 times more efficient to typical GRB spectra (thanks to an improved design of the polarimeter modules)
- Lowered energy threshold to a few keV
- Other payloads being developed in China for joint observations: broad-band spectral imager + low-energy gas polarimeter
- Launch on China Space Station early 2027 (matches LIGO/VIRGO O5 run, possibility of joint observations with GW → **Kole et al. 2023 A&A 669 A77**)





About 20 scientists from 8 institutes:

- UniGe (DPNC), Switzerland: Management, polarimeter modules, instrument thermal and mechanical integration
- UniGe (DA), Switzerland: Online software system
- IAAT, Germany/NCBJ, Poland: Back-End Electronics, Power Supply
- GXU/IHEP, China: Flight Model Acceptance, Spectrometers
- MPE, Germany: Qualification & Verification, Spectrometers
- + INAF-IAPS, Italy

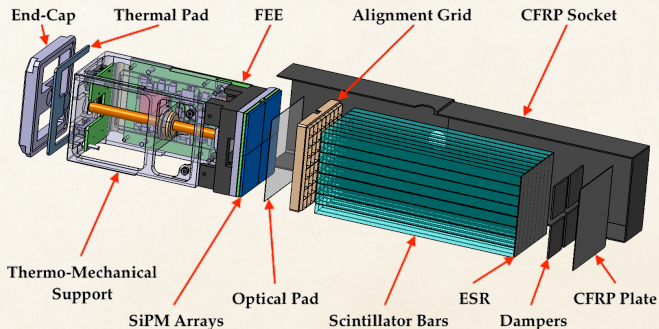


More info on <https://www.unige.ch/dpnc/polar-2>.

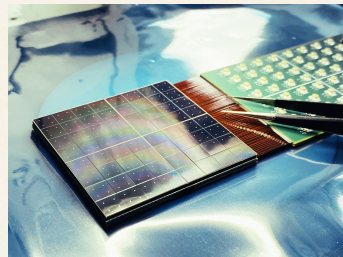




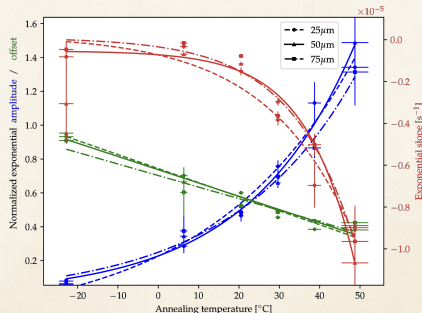
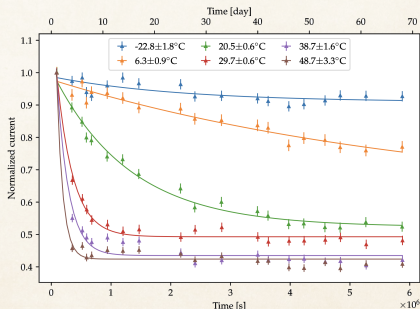
- $8 \times 8$  array of elongated plastic scintillator bars read out by thermally-controlled SiPM arrays
- Many upgrades from the POLAR design:
  - From MAPMTs to SiPM arrays: higher PDE implying a higher effective area and a lowered energy threshold
  - Thinner alignment grid allows to reduce dead-space and increase sensitivity
  - Individual scintillator wrapping + thinner optical coupling pad to improve the optical crosstalk
  - Scintillators have been shortened to improve SNR

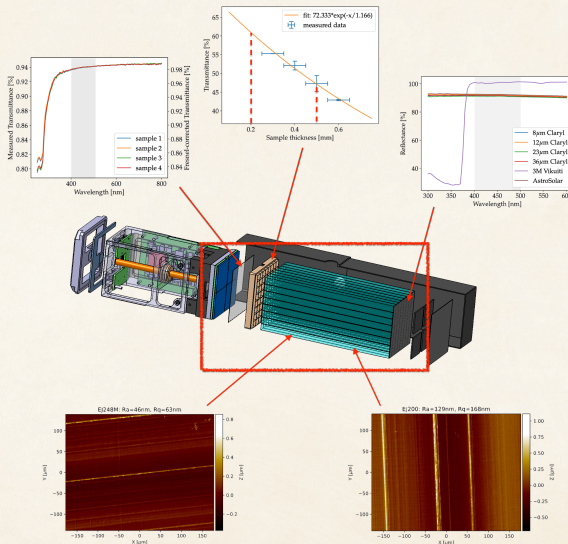
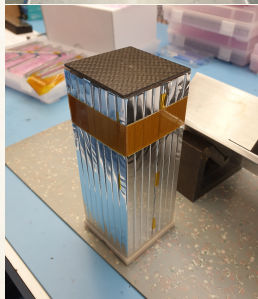
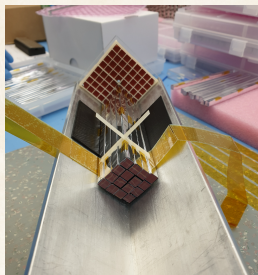


- SiPMs cooled with Peltier elements to reduce dark noise
- Annealing studies allowed to estimate when/how much to heat up the sensors to recover part of the initial performances (degraded due to space radiation environment) → **De Angelis et al. 2023, NIM-A 1048 167934**



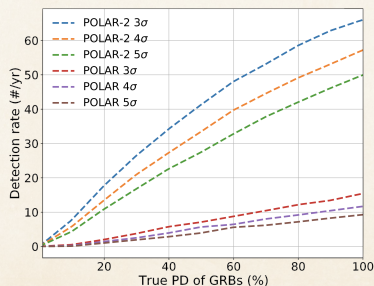
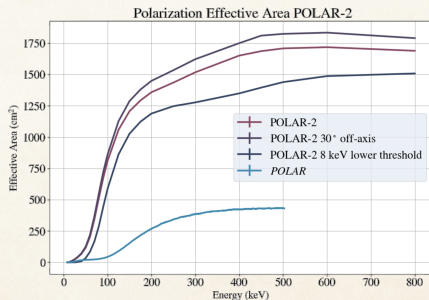
$$a(T) \times \exp(b(T) \cdot t) + c(T)$$



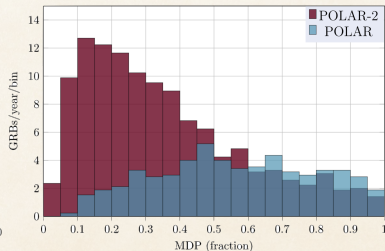
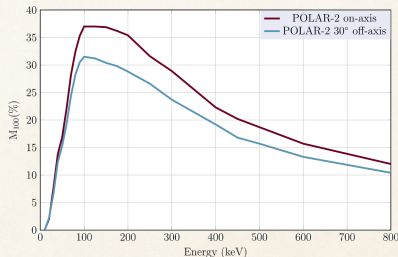


De Angelis et al. 2025, JINST 20 P02010

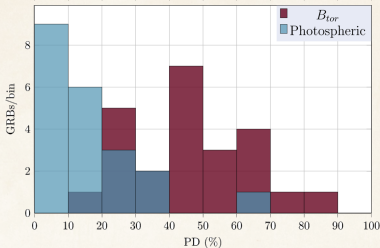
- Module light yield was increased from 0.3 to 1.6 p.e./keV, improving the sensitivity of the instrument and lowering its energy threshold  
→ **De Angelis et al. 2025, JINST 20 P02010**
- The optical crosstalk was also reduced from 15 % down to 1.5%
- Sensitivity to typical GRB spectra improved by an order of magnitude
- Improvement of the modulation factor  $\mu_{100}$ , especially at low energy  
→ **Kole et al. 2024, JINST 19 P08002**



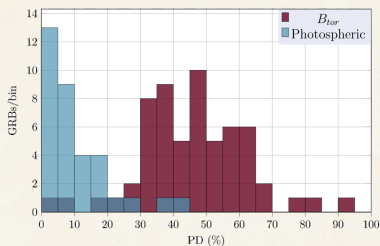
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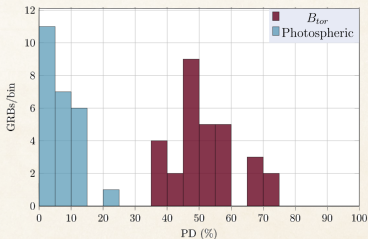
Model Predictions for 1/2 Year of POLAR Observations



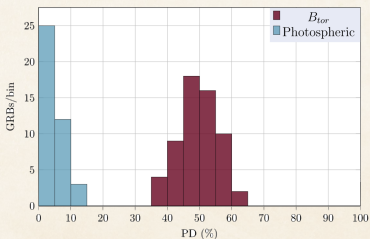
Model Predictions for 1/2 Year of POLAR-2 Observations



Model Predictions for 1/2 Year of POLAR-2 Observations (MPD < 30% cut)



Model Predictions for 2 Years of POLAR-2 Observations (MPD < 15% cut)



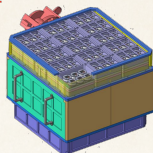


- Two other instruments to enhance the Compton polarimeter capabilities are planned by IHEP, Beijing (**BSD**, imaging spectrometer) and GuangXi University, Nanning (**LPD**, low energy photoelectric polarimeter)

➤ Low-energy Polarization Detector:

LPD

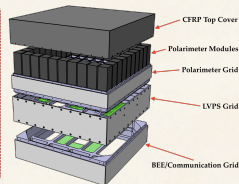
- ~2-10 keV X-ray polarimetry
- Broad energy-band Spectrum Detector: BSD
- ~10-2000keV
- Accurate GRB localization and spectroscopy for HPD and LPD
- Status: Selected, to be adopted



LPD



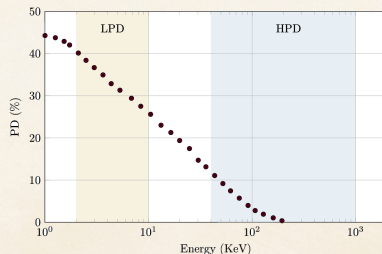
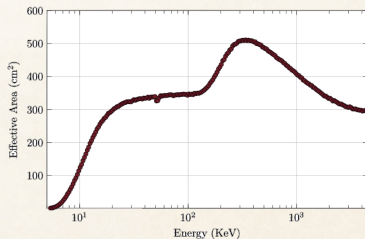
BSD



LPD/GuangXi University

BSD/IHEP

HPD/UniGe



- **POLAR** was **launched in 2016** with the aim to investigate GRB prompt emission polarization, and detected polarization of **14 GRBs**, with a consistency for **low polarization levels (hint for PA temporal evolution)**.
- **POLAR-2**, the successor mission, is now under development with a scheduled **launch mid-2028** to the China Space Station
- Sensitivity to typical GRB spectra was **increased by an order of magnitude** thanks to many design upgrades enhancing the instrumental capabilities
- Sending **alerts to ground** for multi-wavelength follow-up and multi-messenger detection, lifetime matches LIGO O5 Run
- Review in **Kole et al. 2026, Innovations in Space Research Technology**:



*Innovations in Space Research Technology*  
<https://www.scilight.com/journals/SpaceTech>



*Review*

## **Design and Scientific Prospects of the POLAR-2 Mission**

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